

WASEDA University
Graduate School of Information, Production and Systems

Diagnosis-Guided Augmentation and Frequency-Gated Fusion for Actor-Conditioned Human-Value Recognition in News

Peiyang Huang

Midterm Presentation

Data Engineering Lab, IPS, Waseda University

A news topic does not identify an actor's values

Chile protests | Reported government statement

> *"He justified the decision, saying he wanted to maintain* ***public order*** *against vandals and crime."*

Whose conduct?	Which value?	Which direction?
Government and Security Forces	Have a safe country	**Aligned**

Benchmark annotation of reported conduct; not a judgment of permanent character.

The actor, value and direction must be interpreted together.

Source excerpt and gold label: Huang, midterm paper, §5.9, Table 9 (Chile case).

Why this matters for news analysis

20 categories contain 54 fine values

Broad category	Example fine value
Security: societal	Have a safe country
Universalism: concern	Have equality
Self-direction: action	Have freedom of action
Hedonism	Have pleasure

54 values × 2 directions = 108 labels

Challenges for language models

Values may be implicit in an action.

One article can describe several actors.

Intentions and outcomes can conflict.

Measured gap: E1 prompt baseline

1,129 / 2,457 predictions unmatched

1,062 / 2,390 reference labels missed

Exact actor + value + direction match.

One saved run on 177 development articles.

Reliable labels help compare, retrieve and summarize news.

Sources: FEHU value taxonomy; Huang, §§1, 3; E1_REPORT.md, §2 (archived development run).

The task predicts values for each supplied actor

Input

News article

Supplied actor list

Output for each actor

Value + aligned / contradictory

Multiple labels are allowed.

Task	Label space
Task 1b — main task	54 fine values × 2 directions = 108 labels
Task 1a	20 coarse categories, projected from fine values

At test time, Task 1 uses the article—not official subevents.

Task setting: FEHU overview [1]; Huang, midterm paper, §3.1.

Long-tail labels add to actor and direction errors

Implicit evidence

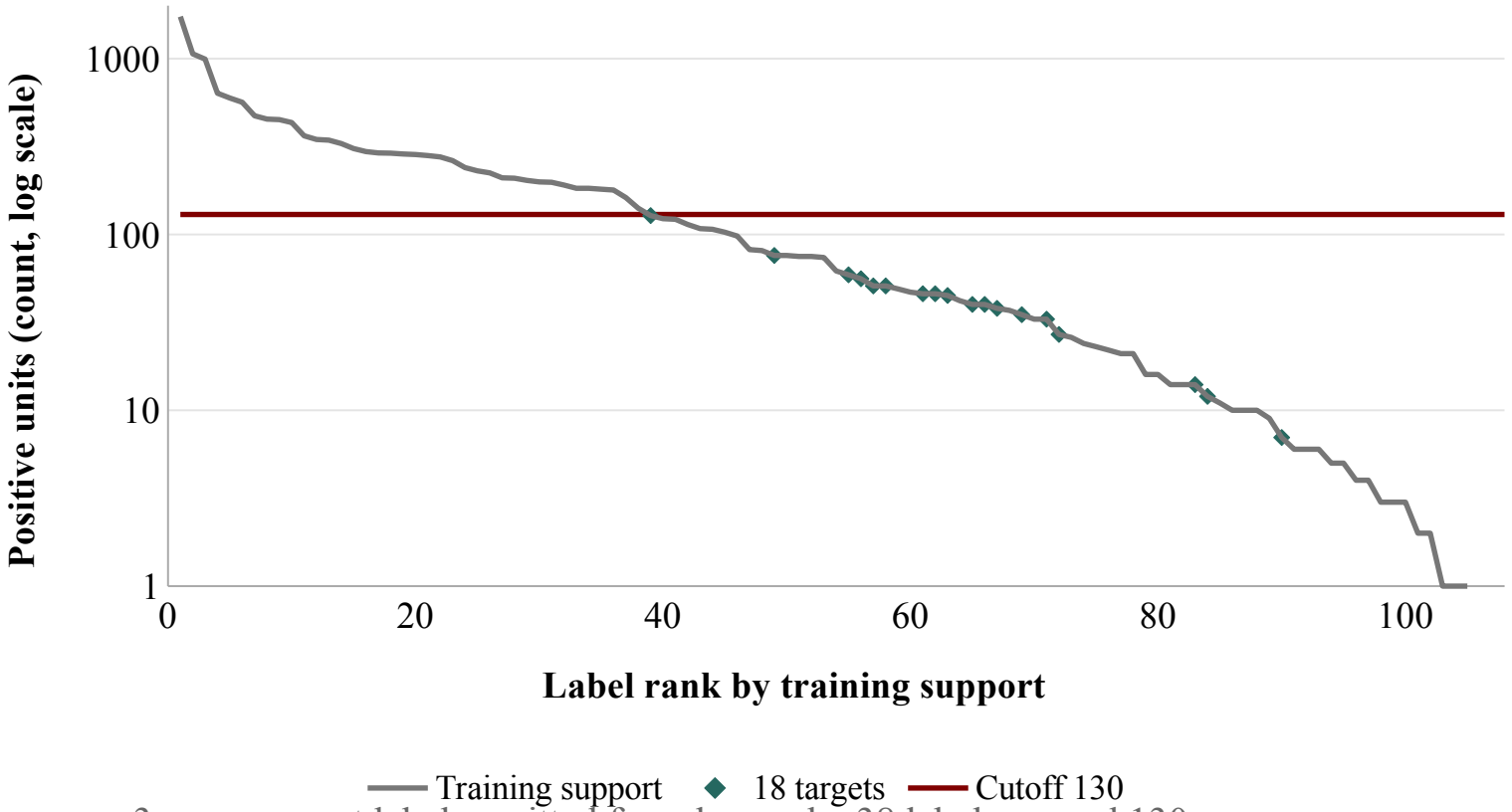
Infer values from reported actions.

Multiple actors

Keep evidence with the correct actor.

Coverage vs errors

Extra labels can add false positives.



3 zero-support labels omitted from log scale; 38 labels exceed 130.

All 18 augmentation targets lie below the removal/addition gate.

Training counts and target ranks: Huang, §5.2, Figure 4. 108 labels; 16,905 positive tuples.

Individual contribution and working hypothesis

Goal: improve actor-specific value recognition
with limited, imbalanced supervision.

My design and analysis	Method	Purpose
DRA-FEHU	Targeted generation + supervision study	Recover missed labels
WAIPS-02	Prompt majority + direction arbitration	Separate two decisions
WAIPS-03 / FGHF	Gated revision + selective direction QA	Control prediction errors

Team boundary: WAIPS-01 belongs to Liu Dongyang.

One frozen target-label prompt from that work supplies auxiliary predictions.

Fusion Micro-F1: 0.5496 → 0.5817. Contribution attribution remains open.

Ideas and progress: Huang, midterm paper, §§1.2–1.3, 4 and Table 7.

Three research questions guide the evaluation

RQ1 Where does recognition fail?

Frequency, representation and direction errors

RQ2 What changes with generated training examples?

Targeted data and unspecified-label supervision

RQ3 How does selective fusion change the errors?

Precision, recall, label balance and direction

Diagnosis → repeated training → saved-stage comparisons

Research questions: Huang, midterm paper, §3.3.

Related work uses a different information flow

Approach	Setting or information flow	Difference here
ValueEval [4]	Human values behind arguments	Actors and directions in news
EAVIT [7]	Local detector → reduced prompt → LLM	Encoder supports prediction revision
This study	**Prompt outputs → gated revision → direction QA**	**Separate selection from direction**

Design comparison; no matched performance experiment against EAVIT.

Encoder support revises prompt predictions before the direction check.

Sources: ValueEval [4], EAVIT [7] (ijcai.org/proceedings/2025/934); Huang, §2.

Two experiment tracks answer different questions

Training study

Generated records → encoder training → three-seed evaluation

Value classifier: DeBERTa-v3-base + LoRA [9, 10]; 108 output scores

Inference study

Prompt predictions + historical encoder scores → fusion stages

The +3.21 pp fusion gain does not isolate the effect of augmentation.

Experimental tracks: Huang, §§1.3, 4.3, 5.4, 5.10 and Appendix A; model choices [9, 10].

Generate training examples for diagnosed failures

METHOD · TARGETED AUGMENTATION

18 target labels: low support and zero baseline recall

Human pilot: 6 records. Hard negatives: 90 requests / 36 explicit negative entries.

Round 2 generation mode	Records (count)
Targeted full examples	450
Direction contrasts	180
Situation transformations	180
Hard-negative requests	90

Round 1: 340 screened records. Round 2: 900 structurally checked records.

Round 2 passed structural checks; semantic label quality remains unverified.

Generation design and record counts: Huang, §4.2, Table 1 and Appendix A.

An unspecified label is not a verified negative

A generated record may describe one target without labeling every other value.

Label status	Assume-negative	Masking
Known positive	Train as positive	Train as positive
Known negative	Train as negative	Train as negative
Unspecified	**Train as negative**	**Exclude from loss**

Masking removes an assumption, but also removes negative supervision.

Supervision comparison: Huang, §§4.3 and 5.6; partial-label learning [8].

Revise eligible labels, then check direction

METHOD · FREQUENCY-GATED FUSION (WAIPS-03)

Recorded model family: DeepSeek v4 Flash / Pro. Exact rules: hidden Appendix, slide 23.

Primary: E6b

Flash predictor
Controlled descriptions
Median: 8 per article

Gated revision

Votes + encoder scores
Calibrate support
Remove / add labels

Direction check

Flash, thinking disabled
Single direction + ≥1
opposite auxiliary vote

Auxiliaries: E1, E6 and Liu target prompt (Flash); E6c (Pro, thinking disabled).

Training support >130: 38 labels. Empirical cutoff chosen during development.

Two-fold article-grouped isotonic calibration. Threshold: primary Micro-F1 / 2 (heuristic).

All 18 targets bypass removal/addition; their direction can still change.

Huang, §§4.4–4.7, Algorithm 1; archived E1/E6/E6b/E6c/E7 reports and QA configuration.

Keep data splits and evaluation measures distinct

Split	Articles (count)	Actors (count)
Train	1,228	3,381
Development	177	459
Task 1 test	175	479

Actors = supplied article–actor units.

Encoder repetitions use seeds 42, 43 and 44.

Micro-F1 ↑

Overall directed-label recognition

Macro-F1 ↑

Equal weight per supported label

Direction reversal rate ↓

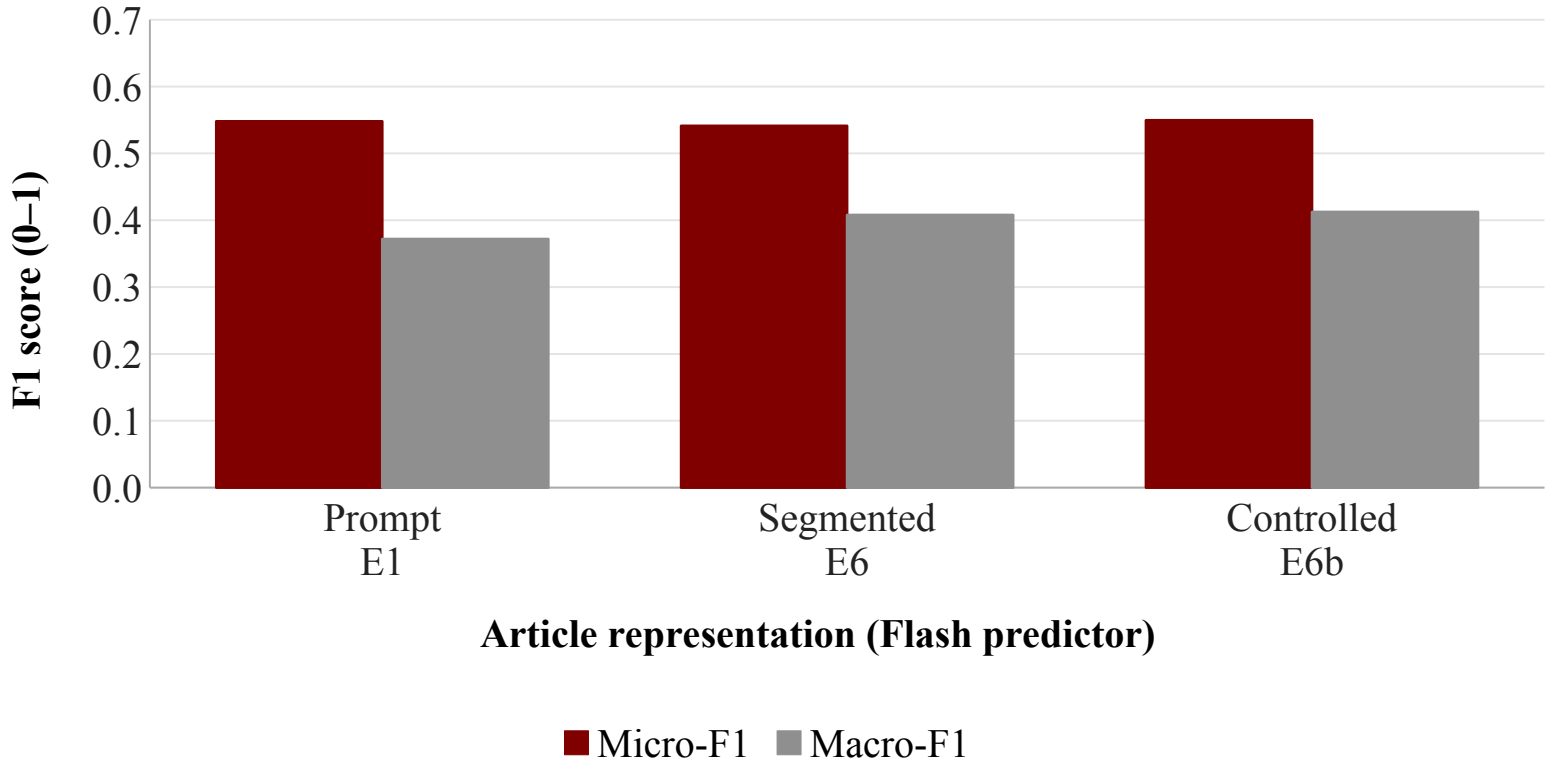
Wrong direction on gold-exclusive pairs

Scores are proportions (0–1). Development choices and official test stay separate.

Dataset counts and metric definitions: Huang, §§5.1–5.3, Table 2.

Representation and model configuration differ

RQ1 · DEVELOPMENT COMPARISONS



Model check: E6c

Same E6b segments

Pro predictor

Thinking disabled

Micro-F1: 0.5343

E6b: 0.5496

Scores: 0–1

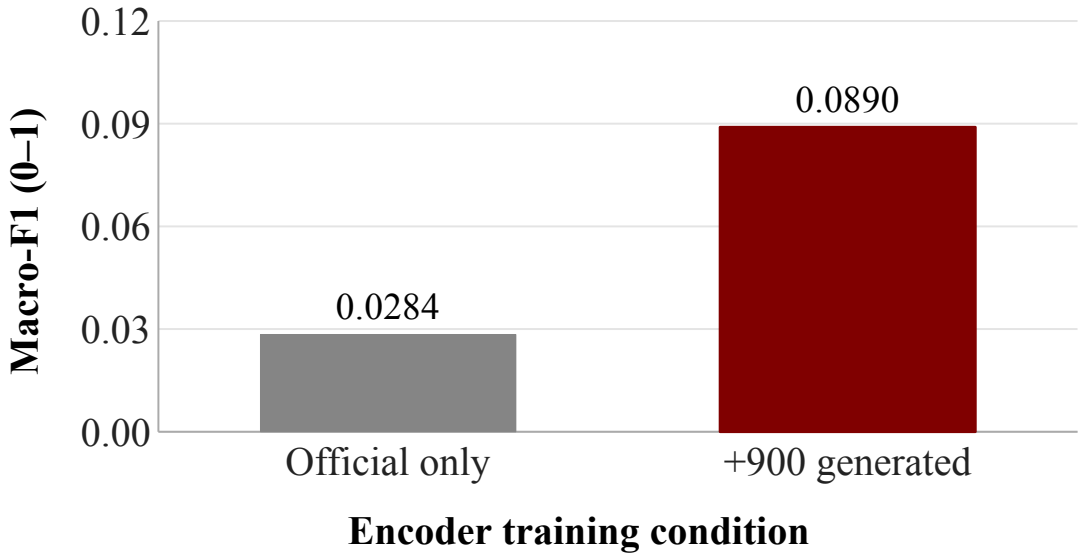
Granularity control helps E6b; the Pro configuration lowers directed F1.

Huang, §5.5, Table 4; E6C_REPORT.md, Design (E6b segments + Pro, thinking disabled).

Ranking improves; target prediction remains weak

RQ2 · AUGMENTATION RESULTS ON DEVELOPMENT

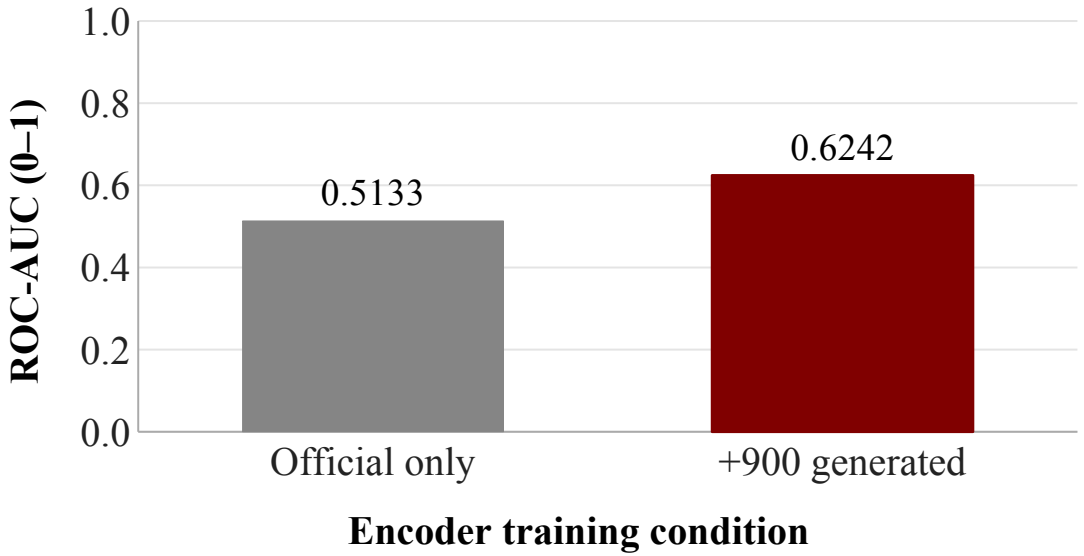
Full-space Macro-F1



Sample SD: 0.0018 / 0.0370

Three-seed means; axis scales differ.

Target ROC-AUC



Sample SD: 0.0304 / 0.0108

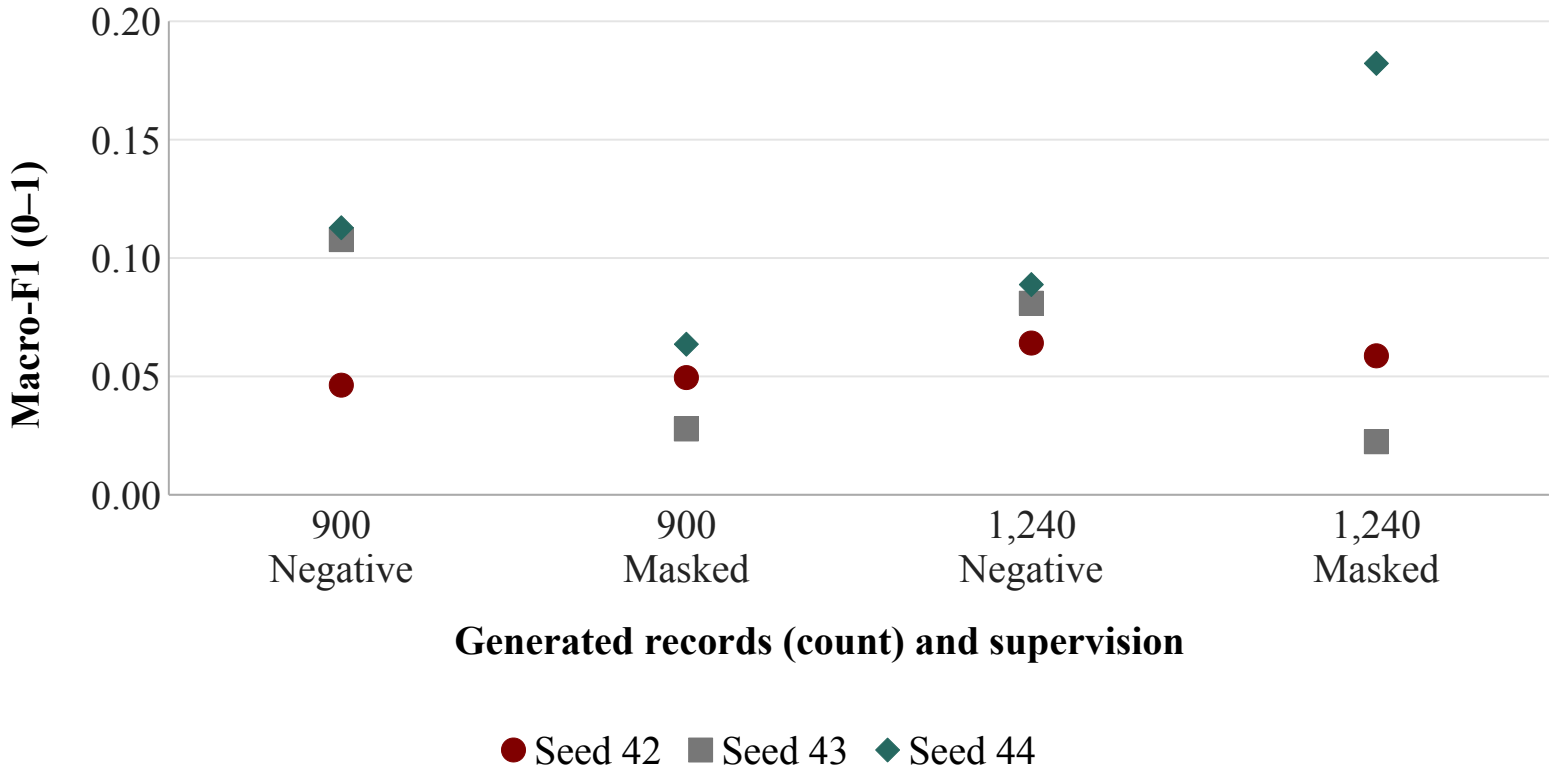
AUC measures ranking, not label accuracy.

Target F1: 0.0000 → 0.0319 at threshold 0.30. Matched data controls are pending.

Huang, §5.6, Table 5 and Figure 5. A vs B: official-only vs 900 assume-negative records.

Masking is sensitive to data and random seed

RQ2 · UNSPECIFIED-LABEL SUPERVISION



Same 900 records

Masking lowers mean Macro-F1.

1,240 masked

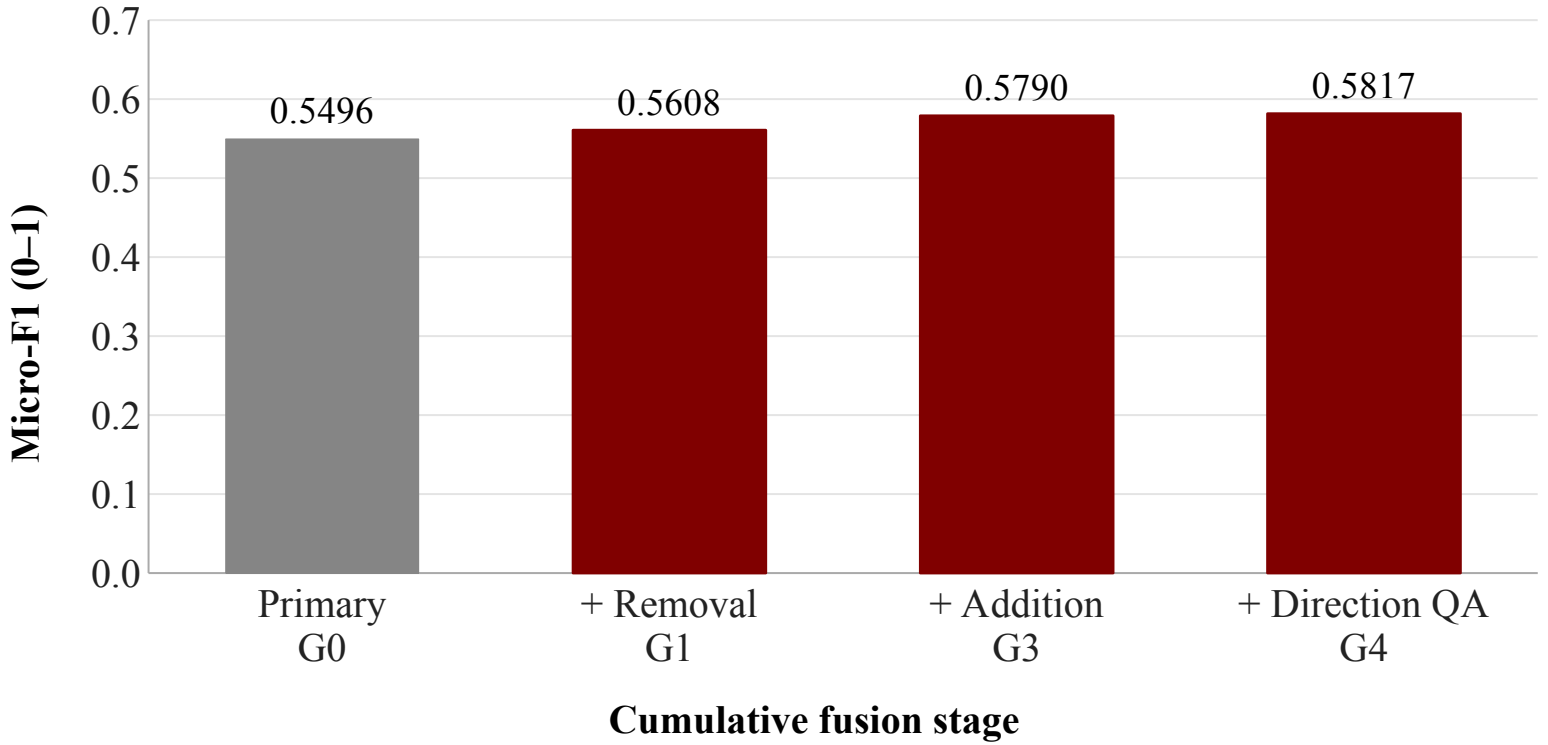
One run drives the higher mean.

Two of three 1,240-record masked runs make no correct target predictions.

Seed values: Huang, §5.6, Figure 5. Negative = treat unspecified labels as negative.

Removal and addition provide most of the gain

RQ3 · FUSION STAGES ON DEVELOPMENT



+3.21 pp

Overall Micro-F1 improvement

95% interval
+2.09 to +4.33 percentage points

Saved outputs only

No correction for model-selection bias

Removal raises precision; addition restores recall. Direction QA adds a small gain.

Huang, §5.7, Table 7 and Figure 6. pp = percentage points; G0 is the fixed primary baseline.

The frequency gate protects label balance

RQ3 · REMOVAL-BRANCH COMPARISON

Removal configuration	Micro-F1 ↑	Macro-F1 ↑
Gated removal (G1)	**0.5608**	**0.4146**
Ungated removal (G5)	0.5703	0.3900

Ungated: +0.95 pp Micro-F1 **Ungated: −2.46 pp Macro-F1**

Scores are proportions (0–1); pp = percentage points.

This is a removal-only comparison, not a complete-system gate ablation.

Huang, §5.7, Table 7. G1 and G5 differ within the removal branch.

Official test results show strengths and limits

EVALUATION · TASK 1b OFFICIAL TEST

Submitted system	Micro-F1 ↑	Macro-F1 ↑	DRR ↓
WAIPS-01 — team comparison	0.5543	0.3277	0.0345
WAIPS-02 — simpler comparator	0.5496	0.3580	0.0395
WAIPS-03 — proposed fusion	**0.5808**	**0.3952**	**0.0477**
IIUoT-01 — other participant	0.5979	0.3930	0.0613
Organizer baseline — unranked	0.6284	0.4402	0.0386

All scores: proportions (0–1). DRR = direction reversal rate. Call budgets differ.

Macro-F1 is 0.22 pp above IIUoT-01. Official ranking uses Micro-F1.

Official scores reproduced from FEHU overview [1] and WAIPS manuscript [2]; Huang, Table 8.

Direction checks can fix or introduce an error

CORRECTED · Venezuela sanctions

*"The situation will **deteriorate** with these last sanctions especially."*

Actor: US Government and Allied Nations

Value: Have a stable society

Aligned → Contradictory

Gold: contradictory

INTRODUCED ERROR · Chile protests

*"...he wanted to maintain **public order** against vandals and crime."*

Actor: Government and Security Forces

Value: Have a safe country

Aligned → Contradictory

Gold: aligned

A label does not show which intention, action or consequence drove the decision.

Excerpts, actor names and gold labels: Huang, §5.9, Table 9. Ellipsis marks a shortened excerpt.

What the current evidence can establish

Supported by the current experiments	Still needs a stronger test
Generated records improve encoder ranking	Targeting vs equal-sized generic data
Saved fusion stages improve the primary set	Augmented encoder contribution in fusion
Direction QA has a small net benefit	Actor-linked evidence and rerun stability

Development informed targets, checkpoints, rules and calibration.

The 18-label list is fixed; the complete 39-to-18 selection rule is not retained.

The system is implemented; contribution attribution is the main remaining task.

Evidence limits: Huang, §§5.10, 6 and Appendix A.

Eight-week plan to test the contributions

FUTURE WORK · PROVISIONAL MILESTONES AFTER MIDTERM FEEDBACK

Proposed effort estimates; sequence and dates will follow supervisor feedback.

Period	Experiment and output	Completion criterion
Weeks 1–2	Encoder ablation Fixed prompts; three encoder conditions	Paired score differences and intervals under one calibration protocol
Weeks 3–4	Equal-size augmentation Targeted / generic / oversampling	Three seeds per condition Target F1 and Macro-F1 comparison
Weeks 5–8	Reliability and evidence Selection split, label audit and reruns	Audit report, rerun spread and reproducible result bundle

Complete the comparisons even if gains are absent; then extend to Task 2.

Planned experiments: Huang, §§6.1–6.2. Relative durations and deliverables are proposed estimates.

Fusion scores, calibration and decision rules



TECHNICAL APPENDIX · AVAILABLE DURING QUESTIONS

Support for one actor–value pair

e: exact directed votes (0–4)

o: opposite-only votes for the same value

p(d): historical encoder probability

Removal: e + 0.25 × o + max(p(0), p(1))

Addition: e + p(d)

Add auxiliary candidates only when their value is absent after removal.

Calibration and constrained decisions

Two pooled isotonic maps: remove / add.

Article hash assigns two folds.

Fit on A, score B; then swap.

For test, fit on all development articles.

Eligible labels: training support >130.

Remove below θ; retain at least 1 tuple.

Add at/above θ; at most 2 tuples per actor.

θ = fitting-fold primary Micro-F1 / 2

Heuristic, not a proven optimum.

Calibration cross-fitting does not remove earlier model-selection bias.

Huang, §§4.6–4.7, Algorithm 1; RH_test/E7_calibrated_jury_fusion/scripts; selection bias [14].

References: task, methods and evaluation

REFERENCES · COMPLETE BIBLIOGRAPHY IN THE NOTES

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Wang et al.; Huang et al., 2026.
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[3, 4] Human values in arguments; ValueEval.
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Local detector + online language model.
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[8] Multi-label learning with partial labels.
Durand et al., CVPR 2019.
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[9, 10] DeBERTaV3; LoRA.
He et al., ICLR 2023; Hu et al., ICLR 2022.
arxiv.org/abs/2111.09543 / 2106.09685

[14] Model selection and selection bias.
Cawley and Talbot, JMLR 2010.
jmlr.org/papers/v11/cawley10a.html

[15] Statistical significance testing in NLP.
Dror et al., ACL 2018.
aclanthology.org/P18-1128/

Complete bibliography: speaker notes.
Citation numbering follows the paper.
School marks: Waseda University / IPS.

Full bibliography: Huang, submitted midterm paper. Sources remain cited on individual slides.

Conclusion: recover values and control the errors

Goal: recognize whose conduct supports or contradicts a value when training supervision is limited and imbalanced.

Contribution	Targeted data, frequency-gated revision and separate direction checking
Main evidence	Development Micro-F1: 0.5496 → 0.5817 Official test: 0.5808 Micro-F1 / 0.3952 Macro-F1
Next step	Isolate the gains with matched comparisons and verify actor-linked evidence

Better label recovery needs careful selection and evidence for direction.

Conclusions: Huang, §§5–6; development Table 7 and official test Table 8.